

CURRENT PROJECT STATUS

Completed

Stage 0

Reverse Engineering

Stage 1

Foundation Refactor

Stage 1.1

Backend Integration

Stage 1.2

Analytics Foundation

Stage 1.3

RBAC Foundation

Current Stage

Stage 1.4

Role Validation & Sync Verification

Must finish before anything else.

STAGE 1.4

Objective

Stabilize platform behavior.

No new features.

No new dashboards.

No redesign work.

Substep 1

Verify User Visibility

Requirements:

User should only see:

Assigned VMs
Personal Analytics
Personal Reports
Settings

Must not see:

Clusters
Nodes
Storage
Global Analytics
Administration

Substep 2

Verify Manager Visibility

Manager should see:

Dashboard
Infrastructure
Virtual Machines
Monitoring
Analytics
Reports
Settings

Must not see:

Administration
Sync Center
User Management
Add VM

Substep 3

Verify Route Protection

Check:

Direct URL access
Sidebar visibility
Protected routes
Role redirects

Substep 4

Verify Sync

Test:

POST /proxmox/cluster/sync

POST /proxmox/nodes/sync

POST /proxmox/storage/sync

POST /proxmox/vms/sync

Document:

Success

Failure

Latency

Errors

Substep 5

Typography Verification

Verify:

Noto Serif

Liberation Serif fallback

Dark mode

Light mode

Tables

Forms

Charts

STAGE 1.4 ANTIGRAVITY PROMPT

Use:

Read:

AI_PROJECT_CONTEXT.md

SYSTEM_REVERSE_ENGINEERING.md

DATABASE_SCHEMA_ANALYSIS.md

API_REFERENCE.md

UI_REDESIGN_ROADMAP.md

Current Stage:

Stage 1.4

Objective:

Validate and stabilize role-based access control, ownership filtering, route protection, synchronization workflows, and typography implementation.

Tasks:

1. Verify User visibility is restricted through vm_user_relation.
2. Verify Manager visibility remains read-only but global.
3. Verify direct URL access protection.
4. Verify sync endpoints function correctly.
5. Add meaningful sync feedback.
6. Verify typography consistency.

Deliver:

- Access matrix
- Sync test report
- Remaining issues
- Files modified

STAGE 2

Dashboard & Infrastructure Center

This is where the application starts becoming impressive.

Objective

Transform infrastructure pages into operational views.

Dashboard

KPI Row

Cards:

Total VMs

Running VMs

Stopped VMs

Nodes
Clusters
Storage Systems

Distribution Charts

VM Status

VMs By Cluster

VMs By Node

Ownership Charts

VMs By Entity

VMs By Division

VMs By Group

Infrastructure Health

Online Nodes

Offline Nodes

Online Storage

Offline Storage

Nodes

Node Summary Cards

Total Nodes

Online

Offline

Total RAM

Total Cores

Node Table

Node
Cluster
IP
Status
Memory
CPU
Uptime
Hypervisor

Visualizations

Memory By Node

CPU By Node

Uptime Comparison

Clusters

Summary

Cluster Count
Node Count
VM Count

Charts

VMs Per Cluster

Storage

Summary

Storage Count
Active Storage
Capacity

Charts

Storage Capacity

Storage Type Distribution

STAGE 2 ANTIGRAVITY PROMPT

Read all project documentation.

Current Stage:

Stage 2

Objective:

Transform Dashboard, Nodes, Clusters, and Storage screens into production-quality operational views.

Requirements:

1. Use only existing APIs.
2. Use only existing schema.
3. No mock data.
4. No placeholder components.

Dashboard:

Implement KPI cards.

Implement VM status chart.

Implement VM distribution charts.

Implement ownership charts.

Implement infrastructure health cards.

Nodes:

Implement node summary cards.

Implement node table.

Implement memory visualization.

Implement CPU visualization.

Clusters:

Implement cluster summary.

Implement VM distribution by cluster.

Storage:

Implement storage summary.

Implement storage capacity charts.

Implement storage type distribution.

Use Plotly.

Use theme-aware charts.

Use Framer Motion.

Provide:

- Screenshots
- Files modified
- APIs used
- Performance considerations

STAGE 3

VM MANAGEMENT REDESIGN

This will likely become the most used screen.

Objective

Transform VM Inventory into a professional infrastructure management interface.

Features

Search

VM Name
VM ID
IP
MAC

Filters

Status
OS
Cluster
Node
GPU

Modern Table

Columns:

Status
VM Name
VM ID
Cluster
Node
CPU
RAM
Disk
IP
GPU

Detail Drawer

Tabs:

Overview
Ownership
Storage
Technical

STAGE 3 ANTIGRAVITY PROMPT

Read all project documentation.

Current Stage:

Stage 3

Objective:

Redesign VM Inventory into the primary operational workspace.

Implement:

Advanced filtering.

Advanced search.

Responsive table.

Right-side detail drawer.

Tabs:

Overview

Ownership

Storage

Technical

Do not alter backend APIs.

Do not alter ownership workflows.

Use existing vmData endpoints.

Provide:

- Screenshots
- Files modified
- Drawer architecture

STAGE 4

MONITORING CENTER

Uses:

/api/stall/cpu

/api/stall/memory

/api/stall/io

Pages

Pressure Analytics

Resource Trends

Charts

CPU Pressure

Memory Pressure

IO Pressure

VM Growth

RAM Growth

Storage Growth

STAGE 4 ANTIGRAVITY PROMPT

Current Stage:

Stage 4

Build Monitoring Center.

Use existing stall APIs.

Implement:

CPU Pressure

Memory Pressure

IO Pressure

Implement:

Node selector

Time selector

Top VM selector

Implement:

Resource growth trends

No fabricated metrics.

Provide:

- API mapping
 - Screenshots
 - Performance notes
-

STAGE 5

ADVANCED ANALYTICS

This is where entity/division insights mature.

Entity Usage

Show:

CPU

RAM

Storage

VM Count

per entity.

Division Usage

Same analysis.

Capacity Projection

Use:

created_date

and historical data.

Only:

Moving Average

Linear Projection

STAGE 5 ANTIGRAVITY PROMPT

Current Stage:

Stage 5

Build Analytics Center.

Implement:

Entity Usage

Division Usage

Capacity Projection

Use only existing schema.

Label forecasts:

Estimated Capacity Trend

Do not claim AI forecasting.

Provide:

- Derivation formulas
 - Charts used
 - Data sources
-

STAGE 6

REPORTS + ADMINISTRATION

Modernize operational workflows.

Reports

Builder

History

Exports

Administration

User Management

Sync Center

Activity Center

STAGE 6 ANTIGRAVITY PROMPT

Current Stage:
Stage 6

Modernize administrative workflows.

Implement:

Report Builder UI

Export UX

Sync Center UX

User Management UX

Improve:
Loading states
Error handling
Success feedback

Do not modify backend contracts.

STAGE 7

USER EXPERIENCE REFINEMENT

Dedicated user-focused portal.

My Dashboard

Only assigned resources.

My VMs

Filtered through:
vm_user_relation

My Analytics

Personal analytics only.

STAGE 7 ANTIGRAVITY PROMPT

Current Stage:

Stage 7

Build dedicated User experience.

All visibility must be filtered through vm_user_relation.

Users must never see:

Clusters

Nodes

Storage

Global Analytics

Provide:

- User journey
 - Screenshots
 - Filtering implementation details
-

STAGE 8

FINAL POLISH

Production readiness.

Tasks

Accessibility

Performance

Skeleton Loaders

Empty States

Error States

Animation Refinement

Cross-Browser Testing

STAGE 8 ANTIGRAVITY PROMPT

Current Stage:

Stage 8

Prepare application for production.

Perform:

Accessibility audit

Performance audit

Animation refinement

Loading state review

Responsive review

Theme consistency review

Deliver:

Final issue list

Final screenshots

Production readiness report

This document, together with the five documentation files, is sufficient to hand the project to another developer or AI and continue development stage-by-stage without losing context.